

Linear systems and elimination

Sharp growth for unequal lower and upper bandwidths

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Lean verification — 22 September 2026

The complete original unequal-bandwidth target is formally proved, strengthened to every $p, q \geq 0$. The sharp growth is $G(0, q) = 1$ and $G(p, q) = h_{p+q}$ for $p > 0$, where $h_t = 0$ for $t \leq 0$ and $h_t = 1 + \sum_{r=1}^p h_{t-r}$ otherwise. The universal upper bound covers every nonsingular complex input with bandwidths at most p, q in its original ordering, every $n \geq 1 + \max(p, q)$, every permitted GEPP tie path, and every intermediate entry. An explicit rational matrix of order $2p + q + 1$ attains the bound along its actual full row-swap path, including both zero-bandwidth endpoints. The entire growth set is nonempty and bounded, has the stated greatest element, and has that genuine real supremum.

All four exports passed LeanCert kernel audits, two independent nonauthor final reviews, and [isolated Linux Comparator/default-kernel verification](#) with actual rejection and sandbox controls. [Proof and reviews](#) · [Immutable proof revision](#) · [Retained evidence](#).

Mathematics: Matthew J. Colbrook; Higham retains original-problem credit. Formalization: Sidney Holden with OpenAI Codex assistance. No normalization, assumed front invariant, real-only restriction, preliminary reordering or deterministic tie rule is substituted. No novelty, source-author endorsement or external human peer review is claimed. The complete original target and earlier source record remain below.

Independently reviewed resolution - 2026-09-11

Sharp growth classification. Theorem 1 and Sections 2-4 prove $G(0, q) = 1$ and $G(p, q) = h_{p+q}$ for $p \geq 1$, where $h_t = 0$ for $t \leq 0$ and $h_t = 1 + \sum_{r=1}^p h_{t-r}$ otherwise. The upper bound covers every complex input and admissible tie path, with growth over all active entries in the fixed original ordering. A real nonsingular matrix of order $2p + q + 1$ attains it, including zero upper bandwidth.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects interacting fill-in and pivot choices across arbitrary unequal bandwidths; specialist impact is a sharp stability classification for banded elimination.

Problem statement

Work in exact arithmetic over $\mathbb{C}$. For integers $p, q \geq 0$, define

$$\mathcal{B}_n(p, q) = \{A \in \mathbb{C}^{n \times n} : \det A \neq 0, \quad a_{ij} = 0 \text{ if } i - j > p \text{ or } j - i > q\}.$$

Gaussian elimination with partial pivoting (GEPP) chooses a largest-modulus entry in the active first column, moves its row to the first position, and forms the trailing Schur complement. For a permitted path π , let $S_1 = A, \dots, S_n$ be its active matrices and define

$$\rho(A, \pi) = \frac{\max_{1 \leq k \leq n} \|S_k\|_{\max}}{\|A\|_{\max}}, \quad \|M\|_{\max} = \max_{i,j} |m_{ij}|.$$

Determine, for all $p \neq q$, the sharp bound independent of dimension,

$$G(p, q) = \sup_{n \geq 1 + \max(p, q)} \sup_{A \in \mathcal{B}_n(p, q)} \sup_{\pi \text{ permitted by GEPP}} \rho(A, \pi).$$

Identify the least universal upper bound and matching examples approaching it. All tie choices are included. The bandwidth restrictions apply in the given ordering; no reordering before GEPP is allowed. Bandwidths are *at most* p and q ; unequal pairs with one zero bandwidth are included. For $p \geq 1$, the equal-bandwidth case is known:

$$G(p, p) = 2^{2p-1} - (p-1)2^{p-2}.$$

This asks how unequal bandwidths affect worst-case element growth in banded linear solves. All unequal pairs constitute one problem.

References

N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, second edition, SIAM (2002), Problem 9.15(a), p. 193; definitions and Theorem 9.11, pp. 172–173. Z. Bohte, *Bounds for Rounding Errors in the Gaussian Elimination for Band Systems*, J. Inst. Maths. Applics. 16 (1975), 133–142.

Earlier status check — 2026-09-08

The book's problem was checked directly. Its wording omits the field; this entry adopts $\mathbb{C}$ from its referenced Theorem 9.11. Searches for *Higham 9.15 growth, growth factor lower bandwidth upper bandwidth*, and *Bohte Gaussian 1975* found no general unequal-bandwidth solution. Shah–Urschel's [August 31, 2026 revision](#), Theorems 2.2–2.3, studies sparsity constraints without determining this extremal function. No recent explicit reaffirmation of this particular question's openness was located; the status rests on the original question and this bounded later-literature check.

Audit update — 2026-09-10

Higham's [Problem 9.15\(a\)](#), p. 193, remains the explicit historical source. Unequal-bandwidth and sharp-growth searches, including the [August 2026 complete-pivoting manuscript](#), did not identify a sharp answer for this partial-pivoting sparsity pattern. This is a bounded historical-source assessment, without a recent explicit reaffirmation.